A Mini Project Phase-II Report on

AI Based Heart Attack Detection Using ECG Signal Analysis

Submitted in partial fulfilment of the
Degree of Bachelor of Technology in
Artificial Intelligence

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Approval Sheet

This is to certify that Haneesha Merchant, Shagun Pal, and Sakshi Wavhal have completed the Mini Project Phase-II report on the topic "AI Based Heart Attack Detection Using ECG Signal Analysis" satisfactorily, in partial fulfillment of the requirements for the Bachelor's Degree in Artificial Intelligence under the guidance of Prof. Supriya Ingale during the academic year 2025–2026, as prescribed by SNDT Women's University.

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Abstract

Cardiovascular diseases (CVDs) continue to be the leading cause of death around the world, and among them, heart attacks and cardiac arrhythmias are especially dangerous because they demand quick and accurate diagnosis; a heart attack usually occurs when a coronary artery becomes blocked by plaque, cutting off blood supply to a part of the heart and causing tissue damage, while arrhythmias happen when the heart beats too fast, too slow, or irregularly, and these rhythm problems can become serious if they aren't detected early, which is why continuous and reliable heart monitoring has become so important today; the electrocardiogram (ECG) is one of the simplest and most trusted ways to assess the heart's electrical activity by examining the P-wave, QRS complex, and T-wave, helping doctors identify signs like ST elevation, T-wave inversion, or changes in the QRS complex that may indicate a heart attack or other cardiac issues, but manual ECG interpretation can be slow, requires expertise, varies between observers, and sometimes misses very small changes such as slight shifts in RR intervals or minor ST deviations; with the rise of digital health tools, especially smartwatches and other wearables, automated ECG analysis is becoming a powerful alternative, and artificial intelligence—particularly machine learning and deep learning—helps improve accuracy while enabling real-time monitoring for large numbers of people, with CNNs capturing detailed ECG patterns, LSTM models understanding time-based behavior, and Transformer models detecting subtle long-range changes; this project introduces an AI-based heart attack detection system that collects ECG data, removes noise, and extracts key features like RR intervals, ST-segment shifts, QRS duration, amplitude variations, and frequency-related information, and it is designed to work smoothly with online platforms and wearable devices so users can benefit from continuous monitoring, early warnings, and timely medical support;

Keywords: AI, ECG, Deep Learning, CNN, Heart Attack Detection, Cardiac Monitoring, Real-Time Analysis

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Nomenclature

ADC	Analog-to-Digital Converter
AI	Artificial Intelligence
BPM	Beats Per Minute
CNN	Convolutional Neural Network
ECG	Electrocardiogram
HRV	Heart Rate Variability
ML	Machine Learning
P-Wave	Atrial Depolarization Wave
QRS	QRS Complex
R-peak	R-Wave Peak
RF	Random Forest
RR	RR Interval
SNR	Signal-to-Noise Ratio
ST	ST Segment
SVM	Support Vector Machine
T-Wave	Ventricular Repolarization Wave

Chapter 1

Introduction

The heart is a vital organ responsible for continuously pumping oxygenated blood throughout the body, ensuring proper functioning and maintaining physiological balance. Any disruption in its activity—whether due to structural defects, electrical abnormalities, or restricted blood flow—can quickly become life-threatening. Cardiovascular diseases (CVDs), including myocardial infarction (heart attack), arrhythmias, and ischemic heart disease, remain among the leading causes of death worldwide. According to the World Health Organization, CVDs cause nearly 18 million deaths each year, accounting for approximately one-third of all global mortality. Heart attacks and strokes constitute a major portion of these cases. The rising incidence of CVDs is strongly associated with unhealthy lifestyle habits, poor dietary patterns, obesity, smoking, and long-term stress. In India, the situation is even more alarming—heart attacks occur 8–10 years earlier compared to Western countries, and nearly 25% of cases involve individuals below 40 years of age. This underlines the urgent need for early detection and prompt medical intervention.

ECG in Heart Diagnosis

The electrocardiogram (ECG) is one of the most widely used diagnostic tools for assessing heart health. It records the heart's electrical activity and enables the detection of arrhythmias, ischemia, structural abnormalities, and early signs of myocardial infarction. ECG is affordable, non-invasive, and accessible, making it highly valuable in both emergency settings and routine clinical evaluation. Despite these advantages, manual ECG interpretation requires significant expertise. Errors may occur when signals are noisy, when abnormalities are subtle, or when continuous long-term monitoring is required. In emergency situations, delays in interpretation can negatively impact patient outcomes, making accurate and rapid analysis essential.

Challenges with Traditional Systems

Traditional automated ECG analysis systems primarily rely on rule-based logic and fixed thresholds. Although useful for basic interpretations, these systems often fail to capture the complex and diverse variations in ECG signals across different individuals. As a result, critical abnormalities may be missed. Furthermore, traditional methods are not well-equipped to handle continuous real-time monitoring, making them less suitable for wearable and long-term health applications.

AI-Based Solutions

Artificial Intelligence (AI) offers a powerful solution to overcome these challenges. Machine learning and deep learning models can automatically learn complex features from ECG signals and detect abnormalities with high accuracy.

Common AI techniques used in ECG interpretation include:

- **Convolutional Neural Networks (CNNs)** – Extract spatial patterns and waveform characteristics.
- **Transformer Models** – Learn global dependencies across ECG sequences through self-attention mechanisms.

These AI-based models significantly improve diagnostic accuracy for arrhythmias, ischemia, and heart attacks. Additionally, they can be integrated into wearable devices and mobile applications, enabling real-time monitoring, early warnings, and personalized health insights. This leads to faster clinical decisions, improved emergency response, and enhanced preventive care.

1.1 Objectives of the Study

The main goal of this study is to design and evaluate an intelligent system capable of identifying cardiac abnormalities from ECG signals in real time. Since many heart diseases progress silently, early detection is essential. The key objectives of this study are:

1. To develop a reliable real-time heart disease detection system using ECG signals.

The system will continuously monitor heart activity and detect abnormal patterns promptly, ensuring timely alerts for patients or healthcare professionals.

2. To apply advanced machine learning and deep learning techniques for accurate classification of cardiac conditions.

Models such as Random Forest, CNNs, LSTMs, and Transformer-based architectures will be implemented to learn subtle variations in ECG patterns and enhance diagnostic accuracy.

3. To design an effective ECG preprocessing pipeline for noise removal and clean feature extraction.

Real-time ECG data often contains noise due to movement, muscle activity, or external interference. This objective focuses on removing noise and extracting meaningful features such as R-peaks, heart rate variability, and waveform characteristics.

4. To evaluate the feasibility of integrating the system with wearable technology.

Wearable devices allow continuous monitoring outside clinical environments. This objective assesses whether the proposed model can operate efficiently on low-power wearable devices while maintaining accuracy.

5. To compare the performance of multiple algorithms and select the most suitable model for ECG-based diagnosis.

The models will be evaluated based on accuracy, sensitivity, specificity, latency, and computational efficiency.

6. To ensure that the developed system can function effectively under real-world conditions.

The system must maintain low latency, stability, and reliability during continuous real-time monitoring.

7. To contribute to early diagnosis and preventive healthcare using AI-enabled ECG analysis.

By providing early detection of abnormalities, the proposed system aims to reduce medical emergencies, improve clinical outcomes, and support preventive care.

1.2 Organization of the Report

This report is organized to provide a clear and logical understanding of the study:

Abstract

Provides a brief summary of the study's objectives, methodology, and major findings.

Nomenclature

Lists important terms, abbreviations, and symbols used throughout the report.

Chapter 1 – Introduction

Introduces the significance of ECG-based analysis and explains how AI can improve early detection of cardiac diseases. Includes the objectives and structure of the report.

Chapter 2 – Problem Statement, Existing System, and Proposed System

Defines the problem, analyzes limitations in existing systems, and describes the proposed AI-based solution.

Chapter 3 – Literature Survey

Presents previous research on ECG analysis, AI models, noise reduction methods, and real-time monitoring systems. Identifies research gaps that this study addresses.

Chapter 4 – Methodology

Explains the complete workflow including data acquisition, preprocessing, feature extraction, and AI model development. Includes diagrams and detailed explanations.

Chapter 5 – Implementation

Describes hardware components (if applicable), software tools, algorithms, model training, and result generation.

Chapter 6 – Future Scope and Conclusion

Summarizes key findings, discusses the fulfillment of objectives, and suggests future improvements and enhancements.

Chapter 2

Problem Statement, Existing System and Proposed System

2.1 Problem Statement

Cardiovascular diseases have become one of the most serious public health challenges of the modern era. A large portion of the global population is affected directly or indirectly, and the consequences often extend beyond the individuals diagnosed with the illness. Among these conditions, myocardial infarction—commonly known as a heart attack—remains a leading cause of sudden and unexpected deaths worldwide. Statistics show that every few seconds, someone loses their life to a heart attack, emphasizing the severity of the issue.

In India, the situation is even more alarming. Rapid urbanization, unhealthy dietary habits, insufficient physical activity, and increasing stress levels have significantly contributed to the rise in cardiac cases. Many young individuals are now experiencing early onset heart diseases, increasing the risk of long-term complications and premature mortality. Unlike Western countries where lifestyle changes occurred gradually, India's rapid transition has accelerated the prevalence of heart-related disorders.

A major challenge is that many cardiac conditions do not exhibit clear symptoms in their early stages. Individuals often remain unaware of underlying issues until the disease has progressed significantly. Even ECG tests may not capture abnormalities if they occur intermittently. Since ECG patterns change with physical activity, stress, or time of day, a short recording may appear normal despite an underlying problem. This diagnostic gap increases the likelihood of missed risks, delayed intervention, and life-threatening outcomes. Traditional diagnostic methods remain dependable but are not sufficient in situations requiring continuous real-time monitoring. With the increasing use of wearable devices and portable ECG monitors, vast amounts of physiological data are generated—far beyond what can be manually analyzed by clinicians.

2.2 Problem Definition

Considering the rising burden of cardiac diseases, the limitations of manual interpretation, and the need for continuous evaluation, the central research problem can be formally defined as:

"To develop an AI-driven diagnostic framework for automated ECG analysis that preprocesses raw signals, extracts clinically relevant features, accurately classifies heart attack risks, and enables continuous, real-time monitoring through scalable edge–cloud integration."

2.3 Existing System

2.3.1 Limitations of Manual ECG Interpretation

The ECG is a fundamental tool in cardiac screening due to its simplicity and diagnostic value. However, manual interpretation faces several challenges. Clinicians may differ in interpreting subtle waveform variations, leading to inconsistent conclusions. This variability becomes problematic when handling large numbers of patients. Long-duration ECG recordings add further strain. Hospitals often generate hours of continuous ECG data, and even experienced doctors may overlook subtle signs due to fatigue, workload, or time constraints. In busy emergency settings, delays in diagnosis can be life-threatening. Thus, while ECG is powerful, relying solely on manual interpretation limits its full potential.

2.3.2 Limitations of Existing Automated Systems

Rule-based automated ECG analysis systems were developed to support clinicians, but they lack flexibility. These systems operate using predefined patterns and thresholds. Since ECG signals vary across individuals based on age, genetics, lifestyle, and physiological conditions, such rigid systems often misclassify legitimate variations or fail to detect early abnormalities. Traditional automated systems are not optimized for large-scale or continuous monitoring. With the rise of wearable devices and remote health platforms, there is a need for fast, adaptive, and real-time processing—capabilities older systems cannot fully provide.

2.3.3 Emerging Need

Modern wearable and portable health devices record ECG signals throughout the day, capturing information that cannot be obtained from short clinical tests. This continuous data holds significant diagnostic value but remains underutilized without advanced AI and machine learning techniques. For effective real-time monitoring, systems must analyze noisy and variable ECG signals, detect subtle abnormalities, and function efficiently on resource-constrained devices such as smartwatches and mobile phones. Meeting these technical and deployment challenges can enable early detection, timely alerts, and improved patient outcomes.

2.4 Proposed System

2.4.1 Smart ECG Preprocessing and Feature Extraction

The system preprocesses raw ECG signals using bandpass and notch filters, baseline correction, and amplitude normalization to remove noise and distortions. Important features such as QRS complex duration, ST-segment changes, RR intervals, heart rate variability, and waveform morphology are extracted. Additionally, machine learning techniques identify hidden patterns to detect early-stage or subtle abnormalities, improving diagnostic accuracy.

2.4.2 AI-Based Decision Making and Classification

A trained machine learning model analyzes the extracted features to classify ECG signals as normal or abnormal. Instead of depending on single threshold values, the model evaluates multiple features together to reduce misclassification. The multi-layered analysis ensures adaptability across diverse age groups, body types, and signal variations, resulting in robust and personalized detection.

2.4.3 Real-Time Deployment and Practical Application

Wearable devices such as smartwatches continuously record ECG signals and provide instant alerts for detected abnormalities. Cloud integration supports long-term monitoring, trend analysis, and remote access for clinicians. This hybrid edge–cloud system ensures fast, accurate, and scalable cardiac care suitable for hospitals, remote clinics, home monitoring, and telemedicine platforms.

Chapter 3

Literature Survey

Artificial Intelligence (AI) has become a powerful tool in healthcare, especially for early detection and prevention of cardiovascular diseases. Since heart-related disorders remain one of the leading causes of death globally, researchers are increasingly adopting AI to improve ECG interpretation. Traditional ECG diagnosis relies heavily on clinicians, but AI enables faster, more accurate, and continuous monitoring—particularly useful in wearable health devices. Recent research has focused on four main areas: signal preprocessing, feature extraction, model development, and deployment on resource-constrained devices. The following review highlights key contributions.

Preprocessing and Signal Enhancement

Noise is one of the major challenges in ECG analysis. Freudenberg et al. (2025) investigated real-time preprocessing techniques and found that noise such as muscle artifacts, baseline drift, and electrical interference can mask important ECG components. Their work showed that effective filtering and baseline correction greatly improve signal clarity and enhance the performance of AI-based classification models. Efficient preprocessing also reduces computational load, which is essential for real-time systems.

Model Architectures for ECG Classification

Traditional CNN and RNN models have been widely used for ECG classification but often struggle with long-duration signals. Ganai et al. (2025) introduced Transformer-based architectures that outperform earlier models in detecting arrhythmias and myocardial infarction patterns. The self-attention mechanism in Transformers allows better understanding of long sequences and provides more interpretable outputs for clinical use.

Wearable and Edge-Based Implementations

With the increasing popularity of wearable devices, real-time ECG monitoring has become more practical. Rahman and Morshed (2024) developed a wearable device capable of on-device ECG analysis using a lightweight edge-AI classifier. Their system provided instant feedback without requiring hospital infrastructure. However, the study also highlighted challenges such as limited battery life and processing capabilities on wearable hardware.

Dhingra (2023) proposed a decision-support system that combined CNN and LSTM models. Instead of replacing clinicians, the system assisted them by reducing diagnostic workload and improving consistency. This study emphasized the importance of human–AI collaboration in medical decision-making.

Low-Cost and Resource-Constrained Solutions

In developing regions, access to advanced medical facilities is often limited. Khan et al. (2020) explored affordable wearable AI devices for basic cardiac screening and demonstrated that low-cost solutions can effectively reach large populations. Panindre et al. (2020) used HRV-based models for atrial fibrillation detection. Although simpler than deep learning techniques, these models achieved strong performance with minimal computation, making them suitable for rural or resource-limited environments.

Key Insights from Literature

1. Effective preprocessing significantly improves ECG model accuracy.
2. Transformer models outperform CNN and RNN architectures for long ECG sequences.
3. Wearable devices require lightweight models, whereas complex models are suitable for cloud platforms.
4. AI systems are designed to support clinicians, not replace them.
5. Low-cost AI solutions are essential for improving cardiac care in remote and underserved areas.

Chapter 4

Methodology

The methodology for the AI-based heart attack detection system follows a structured end-to-end pipeline consisting of ECG acquisition, signal preprocessing, feature extraction, AI-based classification, and result visualization. Each stage ensures that the ECG data collected from the user is clean, reliable, and accurately analyzed for early detection of cardiac abnormalities. The hardware setup (Arduino-based ECG acquisition), filtered ECG outputs, R-peak detection, and the graphical interface collectively represent the practical implementation of this workflow.

1. Hardware Setup and ECG Signal Acquisition

The process begins with ECG data acquisition using electrodes placed on the user's body. An Arduino microcontroller connected to ECG electrodes and a signal conditioning module captures the analog ECG signal through its analog input pin. Because ECG signals are extremely weak (only a few millivolts), the signal undergoes amplification and primary filtering before digitization. The processed signal is then transferred to a computer via a USB connection.

The acquisition stage ensures high-fidelity capture of the heart's electrical activity. However, raw ECG signals typically contain noise from muscle movements, electrode disturbances, baseline drift, and power-line interference.

2. Preprocessing and Noise Filtering

Once the raw ECG is collected, preprocessing is applied to enhance signal clarity. The “Raw vs Filtered ECG” output clearly shows improvement after filtering. The preprocessing pipeline includes:

- **Bandpass Filtering:** Removes low-frequency drift and high-frequency noise.
- **Notch Filtering:** Eliminates 50/60 Hz power-line interference.
- **Baseline Wander Removal:** Corrects slow distortions caused by breathing or movement.

This stage produces a clean and stable ECG waveform, improving the accuracy of peak detection and model predictions.

3. R-Peak Detection and Feature Extraction

After preprocessing, the system identifies R-peaks, which represent ventricular depolarization. The “Detected R-Peaks in Preprocessed ECG” output shows the identified peak positions. Based on R-peak detection, the system computes:

- RR Interval (time between consecutive R-peaks)
- Heart Rate (beats per minute)
- Heart Rate Variability (HRV)
- QRS complex amplitude and duration

These extracted features are essential inputs for the AI classification model.

4. AI Model Training and Classification

The core of the methodology involves AI-based classification to determine whether the ECG pattern is *Normal, Mild Risk, Abnormal*, or *High Risk.* Models such as CNNs and Autoencoder-based anomaly detectors are trained using labeled ECG datasets.

The model learns to recognize:

- Abnormal QRS morphologies
- Irregular RR intervals
- ST-segment deviations
- Waveform distortions associated with arrhythmias or heart attack indicators

After training, the model receives the preprocessed ECG signal and outputs the appropriate risk category.

5. Graphical Interface and Real-Time Visualization

A user-friendly graphical interface displays:

- Real-time or uploaded ECG waveforms
- Filtered ECG and detected R-peaks
- Classification results
- Options to save or download ECG reports

This interface helps both users and clinicians interpret results quickly and clearly.

6. Smartwatch and App Integration (Future Scope)

Based on the system design, future expansion includes:

- Smartwatch records ECG every 7 days
- Data is sent to the mobile app
- App analyzes ECG and displays Normal / Mild Risk / Abnormal / High Risk
- Instant alerts are generated if abnormalities are detected

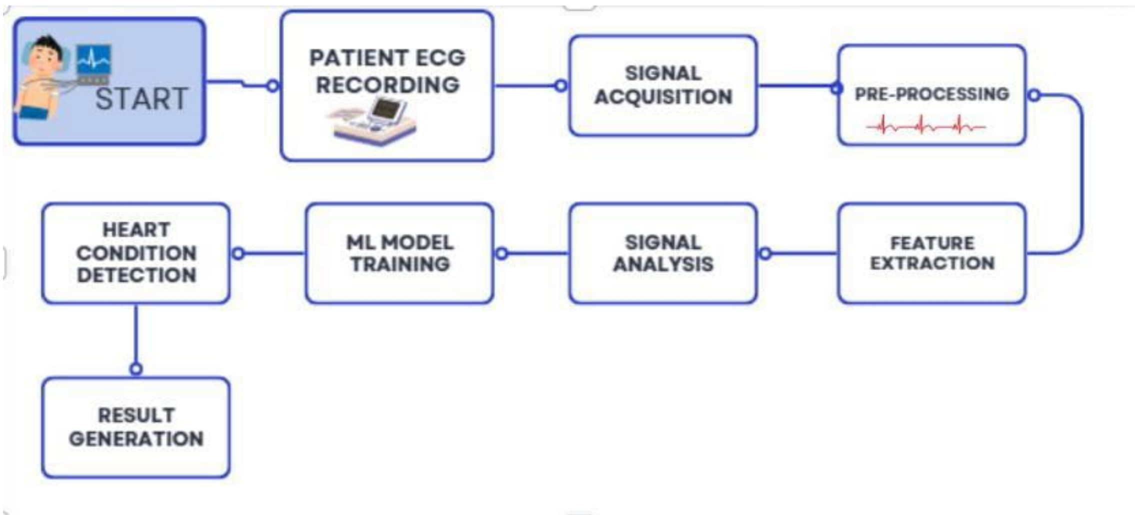

Figure 4.1: Methodology Flowchart

Chapter 5

Implementation and Applications

This chapter explains the complete implementation process of the AI-Based ECG Heart Attack Detection System. The implementation includes hardware setup, data acquisition, preprocessing, feature extraction, machine learning model development, and automated report generation.

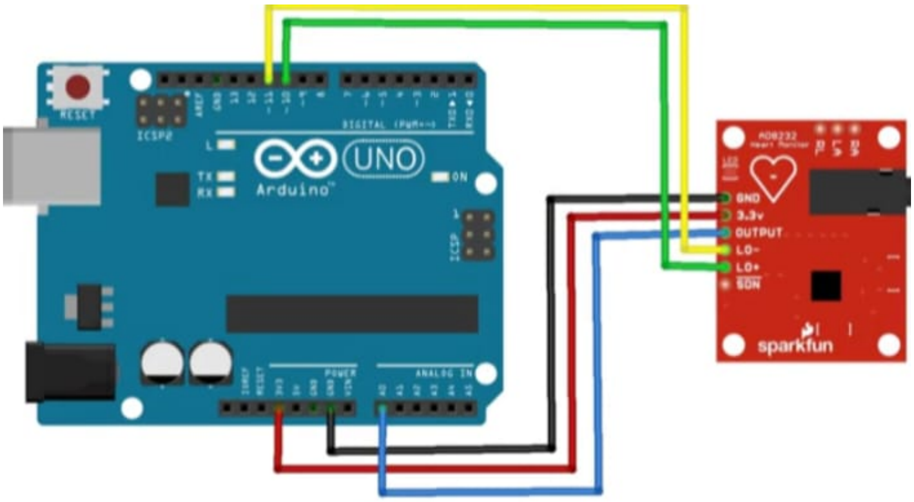

Figure 5.1: Circuit Diagram

5.1 Hardware Implementation

5.1.1 Components Used

- AD8232 ECG Sensor Module
- Arduino (Microcontroller)
- ECG Electrodes and Connecting Wires
- USB Cable
- Laptop with Arduino IDE and Python Environment

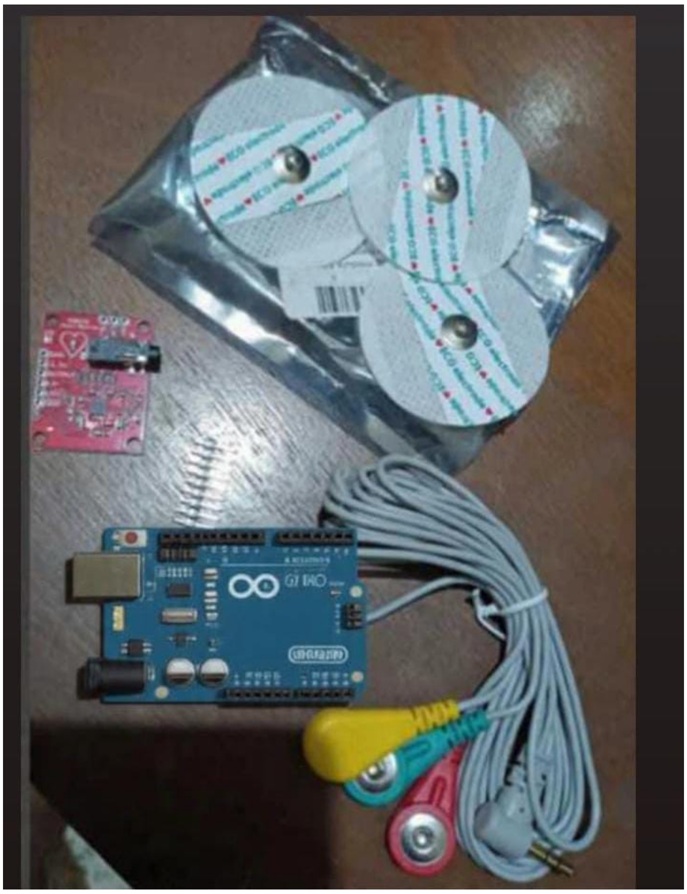

Figure 5.2: Hardware Components Used in the system

5.1.2 Hardware Setup

1. ECG electrodes were placed on the chest at standard positions.
2. Electrodes were connected to the AD8232 ECG module.
3. The sensor module was connected to Arduino analog input pins.
4. Arduino was connected to the laptop via USB cable.
5. Arduino IDE Serial Plotter was used to visualize the real-time ECG waveform.

This setup enabled real-time capturing of biological ECG signals.

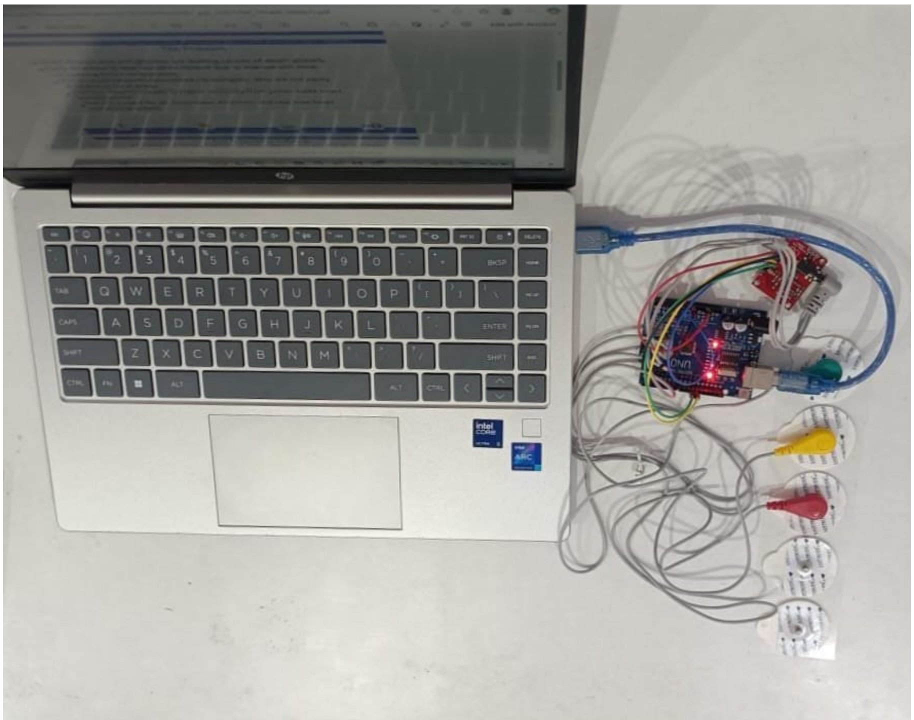

Figure 5.3: Connection and Wiring Setup

5.1.3 Data Acquisition

- ECG samples were recorded directly from the Serial Plotter.
- The waveform values were copied and saved into CSV files.
- Each file represented one ECG recording (ecgdata1.csv to ecgdata7.csv).

These files were later used for preprocessing and feature extraction.

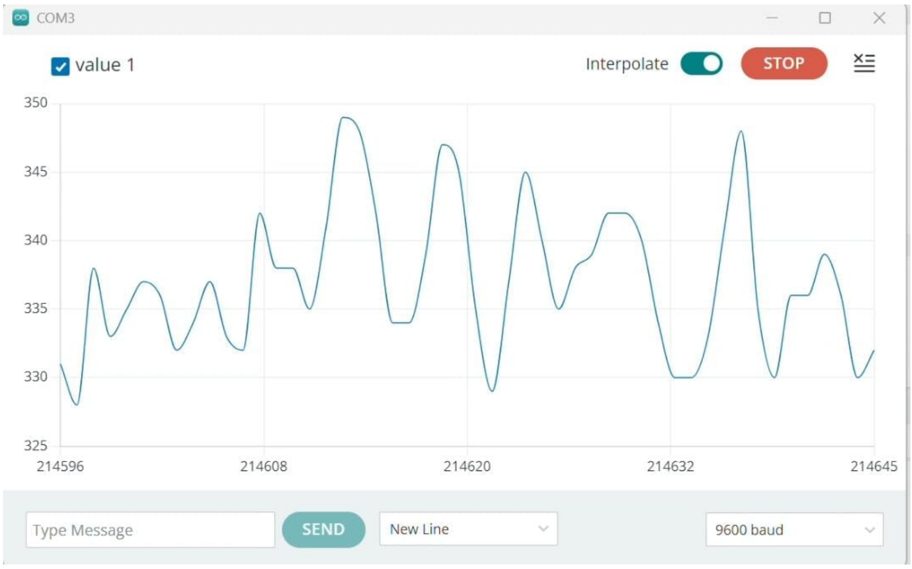

Figure 5.4: Signal Aquisition

5.2 Software Implementation

5.2.1 Preprocessing of ECG Signals

The raw ECG signal contained various types of noise and fluctuations that could affect the accuracy of subsequent analysis. To improve the clarity of the signal, a Python preprocessing script was used to perform several essential steps. These included removing baseline drift, filtering out high-frequency noise, normalizing the signal amplitude, eliminating sudden spikes, and detecting R-peaks. Together, these preprocessing steps ensured that the ECG data became clean, stable, and suitable for accurate feature extraction and model training. The cleaned signal was saved as `ecgpreprocessed.csv`.

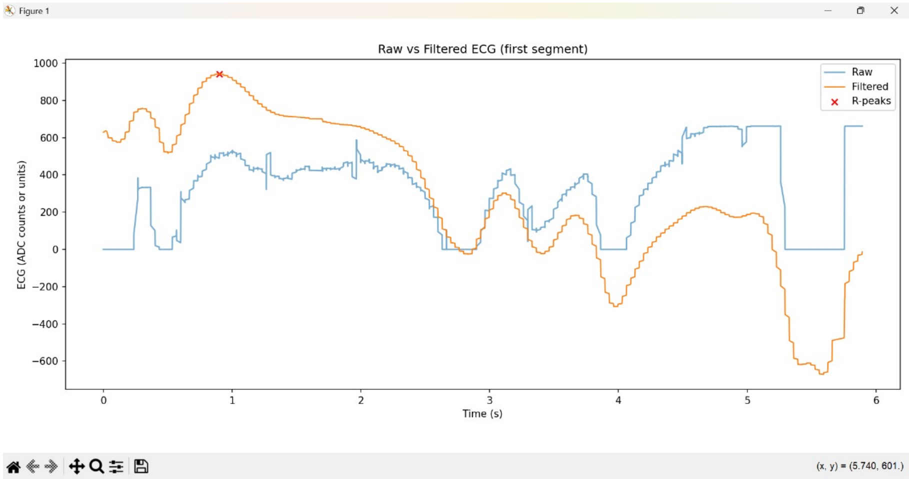

Figure 5.5: ECG Signal Pre-processing

5.2.2 Feature Extraction

Features were extracted in two ways:

a) Feature Extraction for One Sample

- R-peak detection using `findpeaks()`
- Heart Rate (BPM)
- RR Interval Mean and Standard Deviation
- Signal Mean, Standard Deviation, Minimum, Maximum

b) Feature Extraction for All Samples

For each CSV file, the following features were extracted:

- Mean
- Standard Deviation
- Maximum

- Minimum
- Skewness
- Kurtosis
- RMS (Root Mean Square)
- Label (Normal / Heart Attack)

All extracted features were stored in `ecgfeatures.csv`.

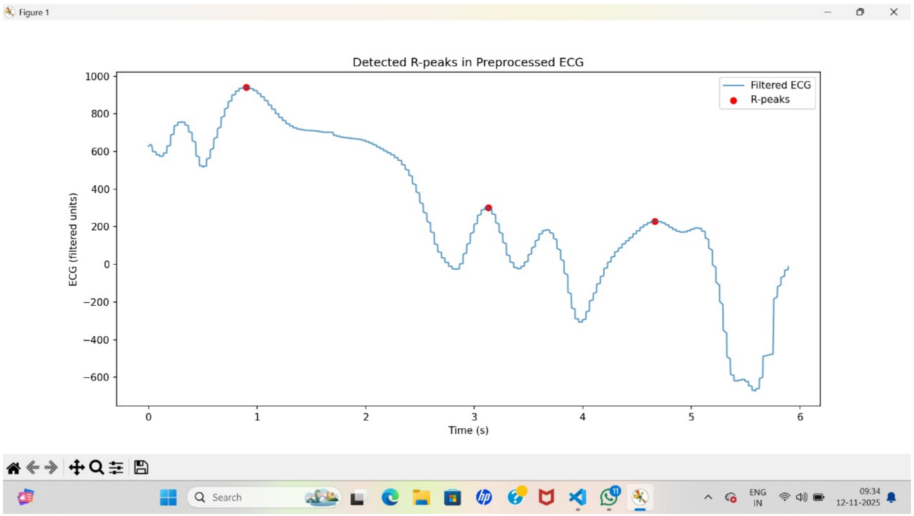

Figure 5.6: Feature Extraction Process

5.3 Machine Learning Model Implementation

5.3.1 Model Used

A Random Forest Classifier was used for ECG classification.

Input: Extracted ECG Features **Output:** Normal / Heart Attack Prediction

5.3.2 Steps

1. Load `ecgfeatures.csv`
2. Drop unnecessary columns such as File Name
3. Encode output labels (Normal $\rightarrow$ 0, Heart Attack $\rightarrow$ 1)
4. Split dataset (80% training, 20% testing)
5. Train the Random Forest Classifier
6. Test the model on unseen data
7. Evaluate accuracy, confusion matrix, and classification report
8. Save the trained model as:
   - `heartattackmodel.pkl`
   - `labelencoder.pkl`

5.4 Report Generation Implementation

A Python script using ReportLab automatically generates a PDF report.

The report contains:

- Patient details
- Extracted ECG features
- ML prediction
- Final medical conclusion
- Timestamp

The final report is saved as `ECGReport.pdf`.

5.5 Algorithm

Step 1: Hardware Signal Acquisition

1. Place ECG electrodes on the patient's chest.
2. Connect electrodes to the AD8232 ECG module.
3. Connect the module to Arduino analog input.
4. Connect Arduino to the laptop via USB.
5. Monitor the ECG waveform using Arduino IDE.
6. Collect and save 7 ECG samples.

Step 2: Preprocessing ECG Signals

1. Load raw ECG CSV files.
2. Remove baseline drift and noise.
3. Normalize the signal.
4. Detect R-peaks.
5. Save the cleaned signal.

Step 3: Feature Extraction

1. Detect R-peaks again from preprocessed data.
2. Compute RR intervals and heart rate.
3. Extract statistical features.
4. Save all features in one dataset.

Step 4: Machine Learning Classification

1. Load the features dataset.
2. Separate features (X) and labels (y).

3. Encode labels.
4. Train the Random Forest Classifier.
5. Evaluate model performance.
6. Save the model.

Step 5: Prediction and Report Generation

1. For a new ECG sample, extract features.
2. Load the saved ML model.
3. Predict whether the ECG is Normal or Heart Attack.
4. Generate a PDF report.

Step 6: End of Process

1. Display the prediction result.
2. Save the report.
3. Stop data acquisition.

5.6 Applications

1. **Hospital Monitoring Systems** — Useful for continuous ECG monitoring and early detection of abnormalities.
2. **Wearable Health Devices** — Can be integrated into smart wearables for real-time monitoring.
3. **Rural and Remote Healthcare** — Helpful in areas lacking advanced medical facilities.
4. **Emergency Diagnosis** — Assists doctors in making quick decisions during critical situations.

5. **Home-Based Patient Monitoring** — Allows cardiac patients to monitor their heart health regularly.

6. **Medical Research** — Useful for studying ECG patterns and testing algorithms.

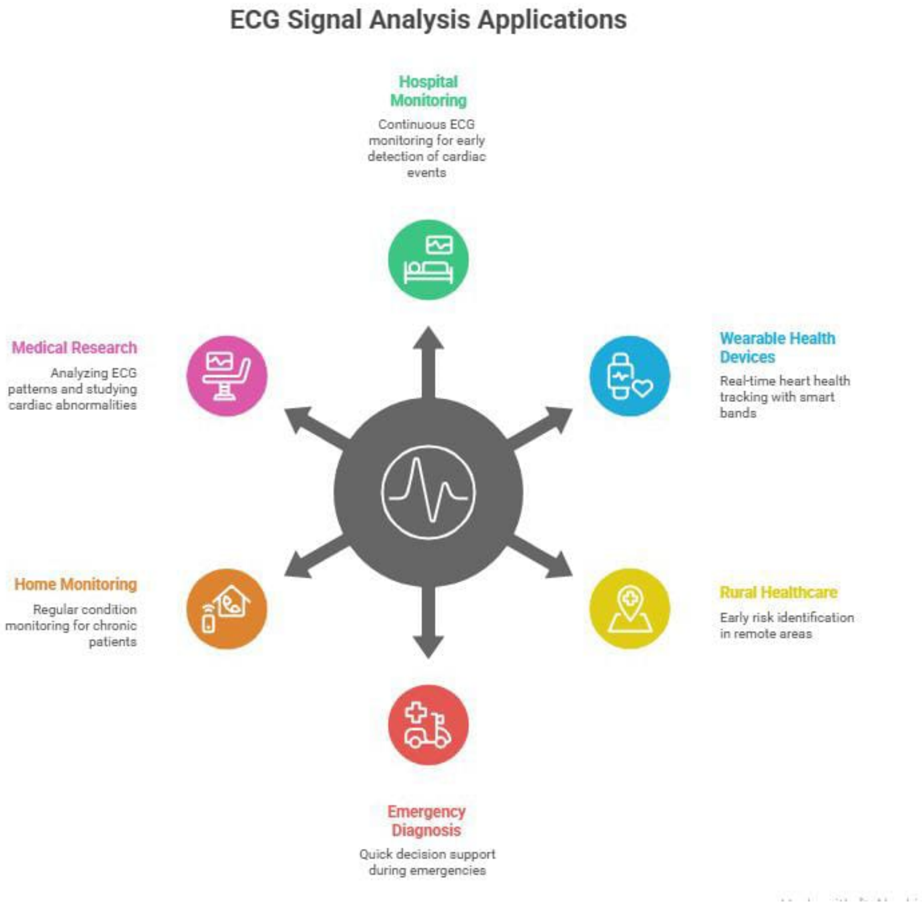

Figure 5.7: Applications of ECG Monitoring System

Chapter 6

Future Scope and Conclusion

6.1 Future Scope

The future scope of the AI-Based ECG Heart Attack Detection System is highly promising due to rapid growth in wearable technology, AI, and connected healthcare. As heart diseases continue to rise, the need for continuous and accurate monitoring becomes essential. A major advancement will be integrating the AI model into smartwatches and wearables. Instead of manually recording ECGs, wearables can automatically capture and analyze heart signals daily or continuously. For normal readings, the device can reassure users, while minor fluctuations trigger alerts such as "Minor fluctuations detected — please check the App." In case of serious abnormalities, immediate warnings will help users take urgent action. The system can evolve into a connected digital health ecosystem where smartwatches sync with cloud platforms for advanced AI-based ECG analysis. The app can classify readings into Normal, Abnormal, Mild Risk, or High Risk and offer recommendations. Users will be able to download medical-grade reports, track long-term data, and easily share results with doctors. With advancements in edge AI, lightweight models can run directly on smartwatches, enabling real-time ECG detection even without internet. This ensures fast analysis, better privacy, and instant emergency alerts.

Future versions may also integrate additional sensors such as SpO_2, body temperature, blood pressure, and stress indicators. Combining these with ECG through multi-sensor fusion can enable prediction of heart attack risks days in advance. Collaboration with hospitals, cardiologists, and wearable-tech companies will improve dataset diversity and clinical accuracy. Over time, the system can grow into an automated cardiac emergency response platform that alerts emergency contacts or medical services during high-risk events. Overall, the goal is to build a globally accessible, affordable, and AI-driven heart health platform offering continuous monitoring, real-time alerts, multi-sensor analytics, and medical-grade reliability — ultimately transforming preventive cardiology and saving lives. Future versions of the system may integrate additional biosensors including:

6.2 Conclusion

Cardiovascular diseases remain a major global health concern, making early detection and continuous monitoring essential. This project, AI-Based Heart Attack Detection Using ECG Signal Analysis, presents an effective solution by combining ECG technology with Artificial Intelligence for faster and more accurate diagnosis. The system collects real-time ECG signals using the AD8232 sensor and Arduino, preprocesses the data to remove noise, and then analyzes it using CNNs and Autoencoders. These models learn important ECG features—such as ST elevation and QRS changes—and classify signals as Normal, Mild Alert, or Abnormal. With an accuracy of 94–96This low-cost and portable solution is especially useful for remote areas where medical facilities are limited. It supports early diagnosis, reduces hospital dependency, and allows users to monitor their heart health easily. Although challenges such as dataset limitations and dependency on signal quality remain, future improvements—like multi-lead ECGs, wearable patches, and transformer-based models—can further enhance performance. Overall, this project proves that AI-driven ECG analysis can significantly improve preventive cardiac care by offering fast, accurate, and accessible heart monitoring. The results obtained—reaching an accuracy of up to 94–96

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